

Assignment

Transformers, Attention Layer, SHAP and LIME

Coursework specification for Transformer modeling and explainability

1. Objective

Build a Transformer-based NLP model, analyze its attention behavior, and explain its predictions using SHAP and LIME. The work should demonstrate both model performance and interpretability on a large dataset.

2. Dataset Requirement

Use the dataset reference provided for Transformer-based NLP experimentation. You may train on a practical subset if hardware resources are limited.

- **Dataset reference:** https://huggingface.co/datasets/fancyzhx/amazon_polarity
- **Instruction:** Follow the reference and choose an appropriate dataset split for your system.

3. Task Breakdown

- **Data preparation:** data collection, cleaning, tokenization, and train/validation/test split.
- **Transformer implementation:** fine-tune BERT/RoBERTa or implement an encoder-based Transformer.
- **Attention analysis:** extract and visualize attention maps from at least two layers and two heads.
- **Explainability:** apply SHAP and LIME to explain individual predictions.
- **Comparative analysis:** compare SHAP and LIME on faithfulness, stability, and runtime.
- **Error analysis:** inspect misclassified samples and explain likely causes.

4. Detailed Requirements

- Report classification metrics: Accuracy, Precision, Recall, and F1-score.
- Include at least 20 explained predictions for SHAP and 20 for LIME.
- Provide at least two attention heatmaps and discuss observed token relations.
- Include at least one flowchart in the report showing your end-to-end methodology.
- Document hardware and software environment used for experiments.
- Ensure code is reproducible with clear run instructions.

5. Deliverables

- **Report (PDF):** 8-12 pages including methods, experiments, results, and conclusion.
- **Code:** notebook or scripts for training, evaluation, attention extraction, SHAP, and LIME.
- **Visual outputs:** attention maps and example explanation plots.
- **README:** setup steps, dataset preparation steps, and commands to reproduce results.
- **GitHub submission:** upload the full project (report, code, and outputs) to a GitHub repository.

6. Evaluation Rubric (100 Marks)

Component	Marks
Data pipeline and large-dataset handling	20
Transformer model design and performance	25
Attention layer analysis quality	20
SHAP and LIME implementation plus comparison	25
Report quality and reproducibility	10

7. Reference Links (Internet Search)

Dataset sources and explainability documentation used to frame this assignment:

- 1) Dataset reference: https://huggingface.co/datasets/fancyzhx/amazon_polarity
- 2) SHAP text examples: https://shap.readthedocs.io/en/latest/text_examples.html
- 3) LIME documentation: <https://lime-ml.readthedocs.io/en/latest/>